

MICHAEL R. HARLEY

Senior Full-Stack Software Engineer · AI/LLM Integration
linkedin.com/in/m-harley | emharley.com | San Francisco, CA

PROFESSIONAL SUMMARY

Full-stack software engineer with 10 years shipping production systems end to end: TypeScript, React, and Next.js on the front, Go and Python on the back, deployed on AWS. Focused over the past few years on putting AI/LLMs into production, including customer-service triage, medical-document generation, and sales automation. Consistent record of measurable impact: \$1.8M saved for FDOT, customer wait time cut from 7 to 2 minutes, 3M records/week processed, and 4+ petabytes of data managed.

TECHNICAL SKILLS

Languages: Go (Golang), Python, TypeScript, JavaScript, C#, SQL (PostgreSQL)

Cloud & Infrastructure: AWS (ECS, S3), Docker, Git, CI/CD, Kafka, MQTT

Frameworks & Tools: Node.js, React.js, Next.js, Django, Flask, FastAPI, REST APIs, Tailwind CSS, Material UI, MongoDB, MariaDB, Elasticsearch, InfluxDB, TimescaleDB

AI / ML: LLMs (ChatGPT, Claude), Machine Learning, TensorFlow, PyTorch, OCR

PROFESSIONAL EXPERIENCE

Pacific Fusion

San Leandro, CA

Senior Full Stack Developer

Starting September 2026

- Building the front-end interfaces and user experiences for the pulser factory hardware testing systems behind Pacific Fusion's inertial fusion Demonstration System.
- Architecting end-to-end data flow for the Demonstration System in React, Go, and Python, integrating multiple diagnostic sources through Kafka/MQTT streaming pipelines.
- Defining database schemas and deploying heterogeneous storage across timeseries (InfluxDB, TimescaleDB), SQL (MariaDB), and NoSQL (MongoDB) systems.
- Partnering with fusion scientists to accelerate the data lifecycle from experiment to analysis.

Deloitte

San Francisco, CA · Remote

Full-Stack Software Engineer

November 2023 – July 2026

- Built a custom dashboard in TypeScript, Next.js, Node.js, and Material UI that ingested and processed 3 million medical records per week, using MongoDB for storage.
- Built a tool in Go, Python, and TypeScript, leveraging AI (ChatGPT), that automatically generated PDF medical documents compliant with Medicare requirements.
- Deployed and operated the medical-records dashboard on AWS ECS, running containerized services in production, with S3-backed storage for 2M+ generated PDFs.
- Led feature development and determined ROI with stakeholders, turning abstract ideas into real-world functionality and integrating AI to increase team productivity.

24 Hour Fitness

San Francisco, CA

Software Engineer – Platform

September 2022 – August 2023

- Integrated ChatGPT with Go, JavaScript, React.js, and Node.js into the customer-service queue to handle high-level product questions, decreasing customer wait time from 7 minutes to 2 minutes.
- Integrated the ActiveCampaign API using JavaScript and Node.js to send 3,000+ customer deals, resulting in \$25,000 in total bookings.

Product Manager

August 2023 – November 2023

- Managed the development and launch of a sales analytics dashboard tracking \$5M+ in transactions and club usage to inform strategic decisions.

Expel, Inc.

Herndon, VA · Remote

Series D/E venture-backed cybersecurity startup (\$1B valuation)

Software Engineer

May 2021 – September 2022

- Implemented and managed the performance of Elasticsearch NoSQL clusters containing over 4 petabytes of data, using Go.
- Created 20+ REST API endpoints in Python and Node.js and integrated 12+ third-party security services, including Microsoft Defender, Cloudflare, and CrowdStrike, to automate workflows.

KPMG

McLean, VA · Hybrid

Software Developer

February 2018 – May 2021

- Built a Python tool that saved the Florida Department of Transportation \$1.8M and 20,000 work hours by automating reconciliation between 600,000 PDF invoices and their financial database.
- Developed a redaction engine using TensorFlow, PyTorch, Python, and UiPath to process 9,000 medical documents and redact PII, reducing manual labor costs by \$250,000.
- Led a team that built a Python application to analyze sentiment and summarize 100,000+ public comments on government regulations, helping policymakers interpret the data.

Symantec

Herndon, VA

Software Engineer

June 2016 – February 2018

- Resolved technical issues in Symantec's log collection platform, used to detect malicious network traffic during the 2016 Rio Olympics.
- Launched an internal web application in Python, Django, and HTML that aggregated data from Symantec's internal databases and external threat-analysis sources.

EDUCATION

American University, College of Arts and Sciences

Washington, D.C.

Bachelor of Science, Computer Science

May 2016